

MTH501 Short Notes Mid term

Lec 1 to18

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Lecture 1 Introduction and Overview

What is Algebra?

Algebra is named in honor of Mohammed Ibn-e- Musa al-Khowârizmî. Around 825, he wrote a book entitled Hisb al-jabr u'l muqubalah, , Al-jabr, presented rules for solving equations.

Algebraic Term

The basic unit of an algebraic expression is a term. In general, a term is either a product of a number and with one or more variables.

Algebraic Expressions

An expression is a collection of numbers, variables, and +ve sign or –ve sign, of operations that must make mathematical and logical behaviour.

For example

$8X^2 + 9X - 1$ is an algebraic expression.

What is Linear Algebra?

One of the most important problems in mathematics is that of solving systems of linear equations. Linear algebra is a branch of mathematics that deals with linear equations and their representations in the vector space using matrices. In other words, linear algebra is the study of linear functions and vectors. It is one of the most central topics of mathematics. Most modern geometrical concepts are based on linear algebra.

Elementary Linear Algebra

Elementary linear algebra introduces students to the basics of linear algebra. This includes simple matrix operations, various computations that can be done on a system of linear equations, and certain aspects of vectors. Some important terms associated with elementary linear algebra are given below:

Scalars

A scalar is a quantity that only has magnitude and not direction. It is an element that is used to define a vector space. In linear algebra, scalars are usually real numbers.

Vectors

A vector is an element in a vector space. It is a quantity that can describe both the direction and magnitude of an element.

Vector Space

The vector space consists of vectors that may be added together and multiplied by scalars.

Matrix

A matrix is a rectangular array wherein the information is organized in the form of rows and columns. Most linear algebra properties can be expressed in terms of a matrix.

Matrix Operations

These are simple arithmetic operations such as addition, subtraction, and multiplication that can be conducted on matrices.

Applications of Linear algebra

Linear algebra makes it possible to work with large arrays of data. It has many applications in many diverse fields, such as

- Computer Graphics,
- Electronics,
- Chemistry,
- Biology,
- Differential Equations,
- Economics,
- Business,
- Psychology,
- Engineering,
- Analytic Geometry,
- Chaos Theory,
- Cryptography,
- Fractal Geometry,
- Game Theory,
- Graph Theory,
- Linear Programming,
- Operations Research

Why using Linear Algebra?

Linear Algebra is then useful for solving problems in such applications in topics such as Physics, Fluid Dynamics, Signal Processing and, more generally Numerical Analysis.

Principal Linear Algebra

There are two principal aspects of linear algebra:

- Theoretical and computational.
- A major part of mastering the subject consists in learning how these two aspects are related and how to move from one to the other.

Recommended Books and Supported Material

The lectures are based on the material taken from the books mentioned below.

1. Linear Algebra and its Applications (3rd Edition) by David C. Lay.
2. Contemporary Linear Algebra by Howard Anton and Robert C. Busby.
3. Introductory Linear Algebra (8th Edition) by Howard Anton and Chris Rorres.
4. Introduction to Linear Algebra (3rd Edition) by L. W. Johnson, R.D. Riess and J.T. Arnold.
5. Linear Algebra (3rd Edition) by S. H. Friedberg, A.J. Insel and L.E. Spence.
6. Introductory Linear Algebra with Applications (6th Edition) by B. Kolman.

Lecture 2

Introduction to Matrices

Matrix

A matrix is a collection of numbers or functions arranged into rows and columns.

Elements

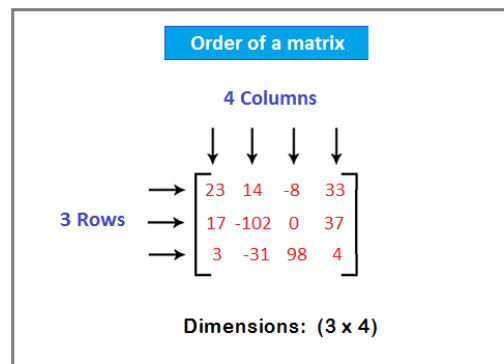
Matrices are denoted by capital letters A, B, K, Y, Z. The numbers or functions are called elements of the matrix. The elements of a matrix are denoted by small letters a, b, K, y, z.

Rows and Columns

The horizontal and vertical lines in a matrix are, respectively, called the rows and columns of the matrix.

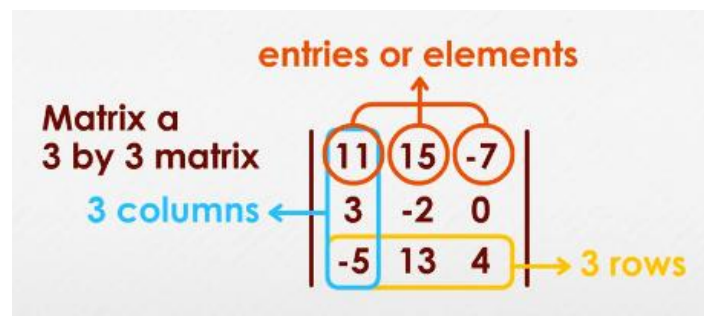
Order of a Matrix

The size (or dimension) of matrix is called as order of matrix. Order of matrix is based on the number of rows and number of columns. It can be written as $r \times c$; r means no. of row and c means no. of columns.



Square Matrix

A matrix with equal number of rows and columns is called square matrix.



Equality of matrices

Two matrices are said to be equal if: Both the matrices are of the same order i.e., they have the same number of rows and columns and $A = B$.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2 \times 2} \quad B = \begin{bmatrix} 6 & 7 \\ 8 & 9 \end{bmatrix}_{2 \times 2}$$

$$A = B$$

$$\Rightarrow \begin{aligned} a &= 6 \\ b &= 7 \\ c &= 8 \\ d &= 9 \end{aligned}$$

Multiple of matrix

A multiple of a matrix A by a nonzero constant k is defined to multiplication of a matrix by a scalar mathematically as: If $A = [a_{ij}]_{m \times n}$ is a matrix and k is a scalar, then kA is another matrix obtained by multiplying each element of A by the scalar k .

$$kA = \begin{bmatrix} ka_{11} & ka_{12} & \cdots & ka_{1n} \\ ka_{21} & ka_{22} & \cdots & ka_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ ka_{m1} & ka_{m2} & \cdots & ka_{mn} \end{bmatrix} = [ka_{ij}]_{m \times n}$$

Addition of Matrices

Only matrices of the same order may be added by adding corresponding elements. If $A = [a_{ij}]$ and $B = [b_{ij}]$ are two $m \times n$ matrices then $A + B = [a_{ij} + b_{ij}]$. Obviously order of the matrix $A + B$ is $m \times n$

$$A = \begin{pmatrix} -4 & 4 & 4 \\ 1 & -4 & 4 \\ -4 & -3 & 4 \\ -5 & 5 & -3 \end{pmatrix}, B = \begin{pmatrix} -5 & 5 & 1 \\ -1 & -5 & -1 \\ 5 & 4 & -3 \\ 2 & -4 & 5 \end{pmatrix}$$

$$A + B = \begin{pmatrix} (-4 + -5) & (4 + 5) & (4 + 1) \\ (1 + -1) & (-4 + -5) & (4 + -1) \\ (-4 + 5) & (-3 + 4) & (4 + -3) \\ (-5 + 2) & (5 + -4) & (-3 + 5) \end{pmatrix}$$

$$= \begin{pmatrix} -9 & 9 & 5 \\ 0 & -9 & 3 \\ 1 & 1 & 1 \\ -3 & 1 & 2 \end{pmatrix}$$

Difference of Matrices

The difference of two matrices A and B of same order $m \times n$ is defined to be the matrix $A - B = A + (-B)$.

Multiplication of Matrices

We can multiply two matrices if and only if, the number of columns in the first matrix equals the number of rows in the second matrix. Otherwise, the product of two matrices is not possible.

$$A_{m \times n} B_{n \times p} = C_{m \times p}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 10 & 11 \\ 20 & 21 \\ 30 & 31 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 10 + 2 \times 20 + 3 \times 30 & 1 \times 11 + 2 \times 21 + 3 \times 31 \\ 4 \times 10 + 5 \times 20 + 6 \times 30 & 4 \times 11 + 5 \times 21 + 6 \times 31 \end{bmatrix}$$

$$= \begin{bmatrix} 10+40+90 & 11+42+93 \\ 40+100+180 & 44+105+186 \end{bmatrix} = \begin{bmatrix} 140 & 146 \\ 320 & 335 \end{bmatrix}$$

Zero Matrix or Null matrix

A matrix whose all entries are zero is called zero matrix or null matrix and it is denoted

by $O = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$;

Associative Law

The matrix multiplication is associative. This means that if A, B and C are $m \times p$, $p \times r$ and $r \times n$ matrices, then $A(BC) = (AB)C$

Distributive Law

If B and C are matrices of order $r \times n$ and A is a matrix of order $m \times r$, then the distributive law states that

- $A(B + C) = AB + AC$
- Furthermore, if the product $(A + B)C$ is defined,
- then $(A + B)C = AC + BC$

Determinant of a Matrix

Associated with every square matrix A of constants, there is a number called the determinant of the matrix,

Which is denoted by $\det(A)$ or $|A|$.

Find the determinant of the following matrix $A = \begin{pmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{pmatrix}$

Solution The determinant of the matrix A is given by

$$\det(A) = \begin{vmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{vmatrix}$$

We expand the $\det(A)$ by first row, we obtain

$$\det(A) = \begin{vmatrix} 3 & 6 & 2 \\ 2 & 5 & 1 \\ -1 & 2 & 4 \end{vmatrix} = 3 \begin{vmatrix} 5 & 1 \\ 2 & 4 \end{vmatrix} - 6 \begin{vmatrix} 2 & 1 \\ -1 & 4 \end{vmatrix} + 2 \begin{vmatrix} 2 & 5 \\ -1 & 2 \end{vmatrix}$$

or

$$\det(A) = 3(20 - 2) - 6(8 + 1) + 2(4 + 5) = 18$$

Transpose of a Matrix

The transpose of $m \times n$ matrix A is denoted by A^T and it is obtained by interchanging rows of A into its columns. In other words, rows of A become the columns of A^T . Clearly A^T is $n \times m$ matrix.

$$\text{If } A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \text{ then } A^T = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}$$

Properties of the Transpose

The following properties are valid for the transpose;

The transpose of the transpose of a matrix is the matrix itself:

The transpose of a matrix times a scalar (k) is equal to the constant times the

transpose of the matrix: $(\underline{ABC})^T = \underline{C}^T \underline{B}^T \underline{A}^T$ $(k\underline{A})^T = k\underline{A}^T$

The transpose of the sum of two matrices is equivalent to the sum of their

transposes: $(\underline{A} + \underline{B})^T = \underline{A}^T + \underline{B}^T$

The transpose of the product of two matrices is equivalent to the product of their

transposes in reversed order: $(\underline{AB})^T = \underline{B}^T \underline{A}^T$

The same is true for the product of multiple matrices: $(\underline{ABC})^T = \underline{C}^T \underline{B}^T \underline{A}^T$

Multiplicative Inverse

Suppose that A is a square matrix of order $n \times n$. If there exists an $n \times n$ matrix B such that $AB = BA = I$, then B is said to be the multiplicative inverse of the matrix A and is denoted by $B = A^{-1}$.

Singular and Non-Singular Matrices

Singular matrix

Singular matrix: A square matrix whose determinant is 0 is called singular matrix.

Nonsingular matrix

Nonsingular matrix: A square matrix that is not singular, i.e. one that has matrix inverse.

Nonsingular matrices are sometimes also called regular matrices. A square matrix is nonsingular iff its determinant is non-zero.

Minor of Matrix

The minor of matrix is for each element of matrix and is equal to the part of the matrix remaining after excluding the row and the column containing that particular element. The new matrix formed with the minors of each element of the given matrix is called the minor of matrix.

The derivative matrix

The definition of differentiability in multivariable calculus is a bit technical. There are subtleties to watch out for, as one has to remember the existence of the derivative is a more stringent condition than the existence of partial derivatives.

$$Df(a) = \left[\frac{df}{dx}(a) \right].$$

$f : \mathbf{R}^n \rightarrow \mathbf{R}$, viewed as a $f(\mathbf{x})$, where $\mathbf{x} = (x_1, x_2, \dots, x_n)$, the $1 \times n$ matrix of partial derivatives at $\mathbf{x} = \mathbf{a}$ is

$$Df(\mathbf{a}) = \left[\frac{\partial f}{\partial x_1}(\mathbf{a}) \quad \frac{\partial f}{\partial x_2}(\mathbf{a}) \quad \dots \quad \frac{\partial f}{\partial x_n}(\mathbf{a}) \right].$$

Lecture 3

Systems of Linear Equations

Linear Equations

Linear equations are equations in which the variables are raised to the power of 1 and are not multiplied or divided by each other. They represent straight lines on a graph and have the general form:

$$y = mx + b$$

Here, y and x are variables, m represents the slope of the line, and b is the y -intercept, which is the point where the line intersects the y -axis.

For example, let's solve the equation $2x + 3 = 9$:

- Start with the equation: $2x + 3 = 9$.
- Simplify: $2x = 9 - 3$, which becomes $2x = 6$.
- Move the constant term (3) to the right side: $2x = 6 - 3$, which becomes $2x = 3$.
- Divide both sides of the equation by 2 to isolate x : $x = 3/2$ or $x = 1.5$.
- Check the solution: $2(1.5) + 3 = 9$. It satisfies the equation.

Homogeneous Linear equation

A system of linear equations having matrix form $AX = O$, where O represents a zero column matrix, is called a homogeneous system.

System of Linear Equations

A finite set of linear equations is called a system of linear equations or linear system. The variables in a linear system are called the unknowns.

General System of Linear Equations

A general system of linear equations consists of multiple equations with multiple variables. It can be represented in matrix form as:

$$A * X = B$$

Here, A is the coefficient matrix, X is the variable matrix, and B is the constant matrix.

Solution of a System of Linear Equations

A solution of a linear system in the unknowns $x_1, x_2, \dots, x_n$ is a sequence of n numbers $s_1, s_2, \dots, s_n$ such that when substituted for $x_1, x_2, \dots, x_n$ respectively, makes every

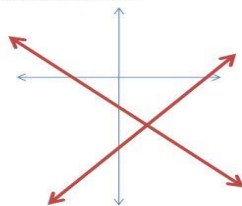
Equation in the system a true statement. The set of all such solutions $\{s_1, s_2, \dots, s_n\}$ of a linear system is called its solution set.

Consistent and inconsistent system

A linear system is said to be consistent if it has at least one solution and it is called inconsistent if it has no solutions.

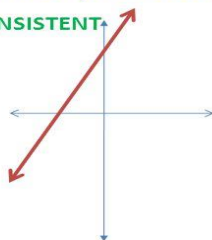
Write the equations of the system below in slope-intercept form.

CONSISTENT



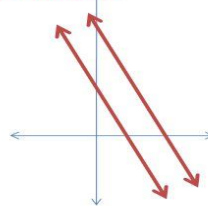
- FIGURE 1.
- $x + y = -2$
- $x - 2y = 7$
- The graphs intersect in exactly one point.

CONSISTENT



- FIGURE 2.
- $y = 2x + 3$
- $2y = 4x + 6$
- The graphs coincide; that is, they have an infinite number of points in common.

INCONSISTENT



- FIGURE 3.
- $3x + y = 2$
- $3x + y = 7$
- The graphs are parallel; that is, they have no points in common.

Parametric Representation

It is very convenient to describe the solution set in this case is to express it parametrically. We can do this by letting $y = t$ and solving for x in terms of t , or by letting $x = t$ and solving for y in terms of t .

Write the parametric equations of a straight line L passing through point $P_1 = (3, 3, -1)$ and the midpoint between $P_2 = (1, -1, 1)$ and $P_3 = (3, 5, 5)$.

(A) $x = 3 - t, y = 3 - t, z = -1 + 4t$

(B) $x = 3 - 2t, y = 3 - t, z = -1 + 4t$

(C) $x = 3 - t, y = 3 - t, z = -1 + 2t$

(D) $x = -2 + 3t, y = 3 - t, z = -1 + 4t$

(E) $x = 3 - 2t, y = 3 - t, z = -1 + 2t$

$$\begin{aligned} x &= x_0 + t\ell \\ y &= y_0 + t\ell^m \\ z &= z_0 + t\ell^n \end{aligned}$$

$$\begin{aligned} x &= 3 - t \\ y &= 3 - t \\ z &= -1 + 4t \end{aligned}$$

direction vector
 $= \langle 2, 2, 3 \rangle - \langle 3, 3, -1 \rangle = \langle -1, -1, 4 \rangle$

midpoint = $(2, 2, 3)$

Matrix Notation

The essential information of a linear system can be recorded compactly in a rectangular array called a matrix.

Given the system

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ -4x_1 + 5x_2 + 9x_3 &= -9 \end{aligned}$$

With the coefficients of each variable aligned in columns, the matrix

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ -4 & 5 & 9 \end{bmatrix}$$

is called the coefficient matrix (or matrix of coefficients) of the system.

$$A_b = \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{bmatrix}$$

It is always denoted by $A_b =$

Elementary Row Operations

1. (Replacement) Replace one row by the sum of itself and a nonzero multiple of another row.
2. (Interchange) Interchange two rows.
3. (Scaling) Multiply all entries in a row by a nonzero constant.

Row equivalent matrices

A matrix B is said to be row equivalent to a matrix A of the same order if B can be obtained from A by performing a finite sequence of elementary row operations of A.

For example $\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & -3 & 13 & -9 \end{bmatrix}$ are row equivalent matrices because we add 4 times of 1st row in 3rd row in 1st matrix.

Lecture 4

Row Reduction and Echelon Forms

Echelon form of a matrix

A rectangular matrix is in echelon form (or row echelon form) if it has the following three properties:

1. All nonzero rows are above any rows of all zeros
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
3. All entries in a column below a leading entry are zero.

Reduced Echelon Form of a matrix

If a matrix in echelon form satisfies the following additional conditions, then it is in reduced echelon form (or reduced row echelon form):

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column

Pivot Positions

A pivot position in a matrix A is a location in A that corresponds to a leading entry in an echelon form of A.

Pivot column

A pivot column is a column of A that contains a pivot position.

Pivot element

A pivot is a nonzero number in a pivot position that is used as needed to create zeros via row operations. The Row Reduction Algorithm consists of four steps, and it produces a matrix in echelon form. A fifth step produces a matrix in reduced echelon form.

Example 3

Apply elementary row operations to transform the following matrix first into echelon form and then into reduced echelon form.

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

Solution

STEP 1

Begin with the leftmost nonzero column. This is a pivot column. The pivot position is at the top

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

└ Pivot column

STEP 2 Select a nonzero entry in the pivot column as a pivot. If necessary, interchange rows to move this entry into the pivot position

Interchange rows 1 and 3. (We could have interchanged rows 1 and 2 instead.)

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

└ Pivot

STEP 3 Use row replacement operations to create zeros in all positions below the pivot

Subtract Row 1 from Row 2, i.e. $R_2 - R_1$

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

└ Pivot

STEP 4 Cover (or ignore) the row containing the pivot position and cover all rows, if any, above it. Apply steps 1–3 to the sub-matrix, which remains. Repeat the process until there are no more nonzero rows to modify.

With row 1 covered, step 1 shows that column 2 is the next pivot column; for step 2, we'll select as a pivot the "top" entry in that column.

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

Pivot
Next pivot column

According to step 3 "All entries in a column below a leading entry are zero". For this subtract 3/2 time R_2 from R_3

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} R_3 - \frac{3}{2}R_2$$

When we cover the row containing the second pivot position for step 4, we are left with a new sub matrix having only one row:

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

Pivot

This is the Echelon form of the matrix.

To change it in reduced echelon form we need to do one more step:

STEP 5 Make the leading entry in each nonzero row 1. Make all other entries of that column to 0.

Divide first Row by 3 and 2nd Row by 2

$$\begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \quad \frac{1}{2}R_2, \quad \frac{1}{3}R_1$$

Multiply second row by 3 and then add in first row.

$$\begin{bmatrix} 1 & 0 & -2 & 3 & 5 & -4 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \quad 3R_2 + R_1$$

Subtract row 3 from row 2, and multiply row 3 by 5 and then subtract it from first row

$$\begin{bmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \quad \begin{matrix} R_2 - R_3 \\ R_1 - 5R_3 \end{matrix}$$

This is the matrix in reduced echelon form.

Lecture 5

Vector Equations

Vector Equations

Vector equations are used to represent the equation of a line or a plane with the help of the variables x, y, z . The vector equation defines the placement of the line or a plane in the three-dimensional framework. The vector equation of a line is $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, and the vector equation of a plane is $\mathbf{r} \cdot \mathbf{n} = d$.

Column Vector

“A matrix with only one column is called column vector or simply a vector”.

$$\text{e.g. } \mathbf{u} = \begin{bmatrix} 3 & -1 \end{bmatrix}^T = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 & 3 & 5 \end{bmatrix}^T = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} w_1 & w_2 & w_3 & w_4 \end{bmatrix}^T \text{ are all}$$

column vectors or simply vectors.

Vectors in $\mathbb{R}^2$

If P is the set of all real numbers then the set of all vectors with two entries is denoted by $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$. The entries in vectors are assumed to be the elements of a set, called a Field. It is denoted by F .

Equal Vector Formula

The formula to check for an equal vector is: If we have two vectors

$\vec{A} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\vec{B} = p\hat{i} + q\hat{j} + r\hat{k}$, And then vectors A and B are equal if and only if $x = p$, $y = q$, and $z = r$, that is, they have equal coordinates.

Equality of vectors in $\mathbb{R}^2$

Two vectors in $\mathbb{R}^2$ are equal if and only if their corresponding entries are equal.

$$\text{If } \mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in \mathbb{R}^2 \text{ then } \mathbf{u} = \mathbf{v} \text{ iff } \boxed{u_1 = v_1} \wedge \boxed{u_2 = v_2}$$

$$\text{So } \begin{bmatrix} 4 \\ 6 \end{bmatrix} \neq \begin{bmatrix} 4 \\ 3 \end{bmatrix} \text{ as } 4 = 4 \text{ but } 6 \neq 3$$

Addition of Vectors

Given two vectors u and v in $\mathbb{R}^2$, their sum is the vector $u + v$ obtained by adding corresponding entries of the vectors u and v , which is again a vector in $\mathbb{R}^2$.

$$\text{For } u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in \mathbb{R}^2 \text{ Then } u + v = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix} \in \mathbb{R}^2$$

$$\text{For example, } \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 1+2 \\ -2+5 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

Multiplication of Vectors and Scalar Quantity

A vector relates two given points. It is a mathematical quantity having both the Magnitude and the direction.

Multiplication of Vectors

Multiplication of vectors can be of two types:

- (i) **Scalar Multiplication**
- (ii) **Vector Multiplication**

Here, we will discuss only the Scalar Multiplication by

Scalar Multiplication of a vector

A vector u and a real number c , the scalar multiple of u by c is the vector cu obtained by multiplying each entry in u by c .

Geometric Descriptions of $\mathbb{R}^2$

Consider a rectangular coordinate system in the plane. Because each point in the plane is determined by an ordered pair of numbers, we can identify a geometric point (a, b) with

the column vector $\begin{bmatrix} a \\ b \end{bmatrix}$. So we may regard $\mathbb{R}^2$ as the set of all points in the plane.

Vectors in $\mathbb{R}^3$

Vectors in $\mathbb{R}^3$ are 3×1 column matrices with three entries. They are represented

geometrically by points in a three-dimensional coordinate space, with arrows from the origin sometimes included for visual clarity.

Vectors in $\mathbb{R}^n$

If n is a positive integer, $\mathbb{R}^n$ (read “ n -space”) denotes the collection of all lists (or ordered n -tuples) of n real numbers, usually written as $n \times 1$ column matrices, such as

$$\mathbf{u} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix}^T$$

The vector whose all entries are zero is called the zero vector and is denoted by $\mathbf{0}$. (The number of entries in $\mathbf{0}$ will be clear from the context.)

Algebraic Properties of $\mathbb{R}^n$

For all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $\mathbb{R}^n$ and all scalars c and d :

- (i) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (Commutative)
- (ii) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (Associative)
- (iii) $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$ (Additive Identity)
- (iv) $\mathbf{u} + (-\mathbf{u}) = (-\mathbf{u}) + \mathbf{u} = \mathbf{0}$ (Additive Inverse)
where $-\mathbf{u}$ denotes $(-1)\mathbf{u}$
- (v) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$ (Scalar Distribution over Vector Addition)
- (vi) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$ (Vector Distribution over Scalar Addition)
- (vii) $c(d\mathbf{u}) = (cd)\mathbf{u}$
- (viii) $1\mathbf{u} = \mathbf{u}$

Linear Combinations

Let V be a vector space over the field K . As usual, we call elements of V *vectors* and call elements of K *scalars*. If $\mathbf{v}_1, \dots, \mathbf{v}_n$ are vectors and $a_1, \dots, a_n$ are scalars, then the *linear combination of those vectors with those scalars as coefficients* is

$$a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + a_3 \mathbf{v}_3 + \cdots + a_n \mathbf{v}_n.$$

Spanning Set

A subset S of a vector space V is called a spanning set for V if $\text{Span}(S) = V$. Examples.

- Matrices $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

form a spanning set for $\mathcal{M}_{2,2}(\mathbb{R})$ as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

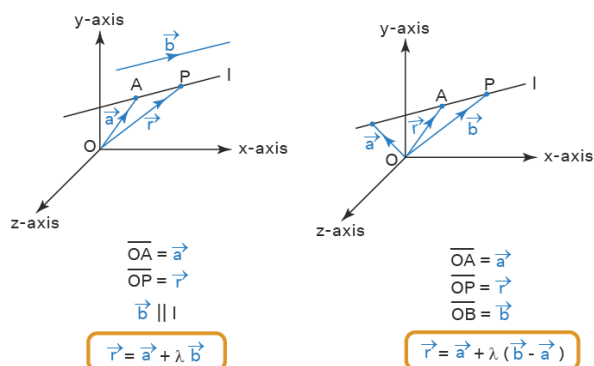
A Geometric Description of Span {u, v}

A Geometric Description of Span {u, v} Take u and v in $\mathbb{R}^3$, with v not a multiple of u. Span {u, v} = plane containing u, v, and the origin 0. = the plane in $\mathbb{R}^3$ spanned by u and v.

Vector Equations of Line

Vector equations of a line can be computed with the help of any two points on the line, or with the help of a point on the line and a parallel vector. The two methods of forming a vector form of the equation of a line are as follows.

Vector Equations Of A Line



- The vector equation of a line passing through a point and having a position vector $\vec{a}$, and parallel to a vector line $\vec{b}$ is $\vec{r} = \vec{a} + \lambda\vec{b}$.
- The vector equation of a line passing through two points with the position vector $\vec{a}$, and $\vec{b}$ is $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$.

So by definition of parallel vectors $\vec{x} - \vec{x}_0 = t\vec{v}$ for some scalar t. it is also called a parameter which varies from $-\infty$ to $+\infty$. The variable point x traces out the line, so the line can be represented by the equation

$$\vec{x} - \vec{x}_0 = t\vec{v} \quad (1) \quad (-\infty < t < +\infty)$$

This is a vector equation of the line through $\vec{x}_0$ and parallel to $\vec{v}$. In the special case, where $\vec{x}_0 = 0$, the line passes through the origin, it simplifies to $\vec{x} = t\vec{v}$ ($-\infty$ to $+\infty$.)

Parametric Equations of a Line in $\mathbb{R}^2$

Let $\vec{x} = (x, y) \in \mathbb{R}^2$ be a general point of the line through $\vec{x}_0 = (x_0, y_0) \in \mathbb{R}^2$ which is parallel to

$v = (a, b) \in R^2$, then eq. 1 takes the form

$$(x, y) - (x_0, y_0) = t(a, b) \quad (-\infty < t < +\infty)$$

$$\Rightarrow (x - x_0, y - y_0) = (ta, tb) \quad (-\infty < t < +\infty)$$

$$\Rightarrow x = x_0 + at, \quad y = y_0 + bt \quad (-\infty < t < +\infty)$$

These are called parametric equations of the line in R^2 .

Lecture 6

Matrix Equations

Matrix Equations

If A is an $m \times n$ matrix, with columns $a_1, a_2, \dots, a_n$ and if x is in R^n , then the product of A and x denoted by Ax , is the linear combination of the columns of A using the corresponding entries in x as weights, that is,

$$Ax = \begin{bmatrix} a_1 & a_2 & \dots & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 a_1 + x_2 a_2 + \dots + x_n a_n$$

Which has the form $Ax = b$, and we shall call such an equation a matrix equation, to distinguish it from a vector equation.

When we say “ A is an $m \times n$ matrix,” we mean that A has m rows and n columns.

Lecture 7

Solution Sets of Linear Systems

Solution Set

A solution of a linear system is an assignment of values to the variables $x_1, x_2, \dots, x_n$ such that each of the equations in the linear system is satisfied. The set of all possible solutions is called the Solution Set.

Homogeneous Linear System

A system of linear equations is said to be homogeneous if it can be written in the form $Ax = 0$, where A is an $m \times n$ matrix and 0 is the zero vector in $\mathbb{R}^m$.

Trivial Solution

A homogeneous system $Ax = 0$ always has at least one solution, namely, $x = 0$ (the zero vector in $\mathbb{R}^n$). This zero solution is usually called the trivial solution of the homogeneous system.

Nontrivial solution

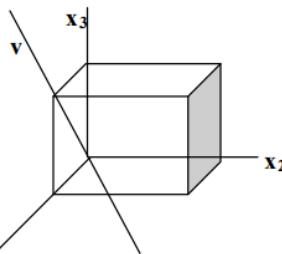
A solution of a linear system other than trivial is called its nontrivial solution. i.e the solution of a homogenous equation $Ax = 0$ such that $x \neq 0$ is called nontrivial solution, that is, a nonzero vector x that satisfies $Ax = 0$.

Existence and Uniqueness Theorem

The homogeneous equation $Ax = 0$ has a nontrivial solution if and only if the equation has at least one free variable.

Geometric Interpretation

Since neither u nor v is a scalar multiple of the other, so these are not parallel, the



solution set is a plane through the origin.

Parametric Vector Form of the solution

The equation $x = su + tv$ (s, t in $\mathbb{R}$) is called a parametric vector equation of the plane.

Steps of Writing a Solution Set (of a Consistent System) in a Parametric Vector Form

- Step 1: Row reduces the augmented matrix to reduced echelon form.
- Step 2: Express each basic variable in terms of any free variables appearing in an equation.
- Step 3: Write a typical solution x as a vector whose entries depend on the free variables if any.

- Step 4: Decompose x into a linear combination of vectors (with numeric entries) using the free variables as parameters.

Lecture 8

Linear Independence

Linear Independence

An indexed set of vectors $\{v_1, v_2, \dots, v_p\}$ in $\mathbb{R}^n$ is said to be linearly independent if the vector equation $x_1v_1 + x_2v_2 + \dots + x_pv_p = \mathbf{0}$ has only the trivial solution. An indexed set of vectors $\{v_1, v_2, \dots, v_p\}$ is said to be linearly dependent if there exist weights $c_1, \dots, c_p$, not all zero, such that $c_1v_1 + c_2v_2 + \dots + c_pv_p = \mathbf{0}$.

Linear Independence of Matrix Columns

Suppose that we begin with a matrix $A = [a_1 \ \dots \ a_n]$ instead of a set of vectors. The matrix equation $Ax = \mathbf{0}$ can be written as $x_1a_1 + x_2a_2 + \dots + x_na_n = \mathbf{0}$.

Each linear dependence relation among the columns of A corresponds to a nontrivial solution of $Ax = \mathbf{0}$.

The columns of a matrix A are linearly independent if and only if the equation $Ax = \mathbf{0}$ has only the trivial solution.

Determine whether the columns of $A = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & -1 \\ 5 & 8 & 0 \end{bmatrix}$ are linearly

independent.

Solution To study $Ax = \mathbf{0}$, row reduce the augmented matrix:

$$\begin{array}{l} \begin{bmatrix} 0 & 1 & 4 & 0 \\ 1 & 2 & -1 & 0 \\ 5 & 8 & 0 & 0 \end{bmatrix} \\ \sim \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 5 & 8 & 0 & 0 \end{bmatrix} \\ \sim \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & -2 & 5 & 0 \end{bmatrix} \\ \sim \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 13 & 0 \end{bmatrix} \end{array} \quad \begin{array}{l} \\ R_{12} \\ (-5)R_1 + R_3 \\ (2)R_2 + R_3 \end{array}$$

Lecture 9

Linear Transformations

Matrix Equation

An equation $Ax = b$ is called a matrix equation in which a matrix A acts on a vector x by multiplication to produce a new vector called b .

$$\begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

$\uparrow \qquad \qquad \uparrow \qquad \qquad \uparrow$
 $A \qquad \qquad x \qquad \qquad b$

and

$$\begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\uparrow \qquad \qquad \uparrow \qquad \qquad \uparrow$
 $A \qquad \qquad u \qquad \qquad \underline{0}$

Solution of Matrix Equation

Solution of the $Ax = b$ consists of those vectors x in the domain that are transformed into the vector b in range. Matrix equation $Ax = b$ is an important.

Transformation of Function or Mapping

A transformation (or function or mapping) T from R^n to R^m is a rule that assigns to each vector x in R^n , an image vector $T(x)$ in R^m .

$$T : R^n \rightarrow R^m$$

The set R^n is called the domain of T , and R^m is called the co-domain of T . For x in R^n the set of all images $T(x)$ is called the range of T .

Shear transformation

The transformation $T : R^n \rightarrow R^m$ defined by $T(x) = Ax$ is called a shear transformation.

Linear transformations

Linear transformations preserve the operations of vector addition and scalar multiplication

Properties

If T is a linear transformation, then

1. $T(\mathbf{0}) = \mathbf{0}$
2. $T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$
3. $T(c_1\mathbf{v}_1 + \dots + c_p\mathbf{v}_p) = c_1T(\mathbf{v}_1) + \dots + c_pT(\mathbf{v}_p)$

for all vectors $\mathbf{u}, \mathbf{v}$ in the domain of T and all scalars c, d .

Example Define a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$T(\mathbf{x}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$$

Find the images under T of $\mathbf{u} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, and $\mathbf{u} + \mathbf{v} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$.

Solution $T(\mathbf{u}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \quad T(\mathbf{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$

$$T(\mathbf{u} + \mathbf{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}$$

In the above example, T rotates $\mathbf{u}, \mathbf{v}$ and $\mathbf{u} + \mathbf{v}$ counterclockwise through 90° .

In fact, T transforms the entire parallelogram determined by $\mathbf{u}$ and $\mathbf{v}$ into the one determined by $T(\mathbf{u})$ and $T(\mathbf{v})$

Example Let $L: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation for which:

$$L(1, 0, 0) = (2, -1),$$

$$L(0, 1, 0) = (3, 1), \text{ and}$$

$$L(0, 0, 1) = (-1, 2).$$

Then find $L(-3, 4, 2)$.

Solution Since $(-3, 4, 2) = -3\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$,

$$\begin{aligned} L(-3, 4, 2) &= L(-3\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}) = -3L(\mathbf{i}) + 4L(\mathbf{j}) + 2L(\mathbf{k}) \\ &= -3(2, -1) + 4(3, 1) + 2(-1, 2) = (4, 11) \end{aligned}$$

Lecture 10

The Matrix of a Linear Transformation

Theorem

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that $T(x) = Ax$ for all x in $\mathbb{R}^n$.

In fact, A is the $m \times n$ matrix whose j^{th} column is the vector $T(e_j)$, where e_j is the j^{th} column of the identity matrix in $\mathbb{R}^n$.

$$A = [T(e_1) \quad \dots \quad T(e_n)]$$

Proof Write

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ x_2 \\ \vdots \\ 0 \end{bmatrix} + \dots + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ x_n \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \dots + x_n \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

$$= x_1 e_1 + \dots + x_n e_n = \begin{bmatrix} e_1 & \dots & e_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} e_1 & \dots & e_n \end{bmatrix} x$$

Since T is Linear, So

$$T(x) = T(x_1 e_1 + \dots + x_n e_n) = x_1 T(e_1) + \dots + x_n T(e_n)$$

$$= \begin{bmatrix} T(e_1) & \dots & T(e_n) \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = Ax$$

The matrix A in (1) is called the standard matrix for the linear transformation T . We know that every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a matrix transformation and vice versa.

Existence and Uniqueness of the solution of $T(x)=b$

Existence

A mapping: $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be onto $\mathbb{R}^m$ if each b in $\mathbb{R}^m$ is the image of at least one x in $\mathbb{R}^n$.

Uniqueness

A mapping: $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be one-to-one if each b in $\mathbb{R}^m$ is the image of at most one x in $\mathbb{R}^n$.

Theorem

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then T is one-to-one if and only if the equation $T(x) = 0$ has only the trivial solution.

Proof:

Since T is linear, $T(0) = 0$. If T is one-to-one, then the equation $T(x) = 0$ has at most one solution and hence only the trivial solution. If T is not one-to-one, then there is a b that is the image of at least two different vectors in $\mathbb{R}^n$ (say, u and v). That is, $T(u) = b$ and $T(v) = b$.

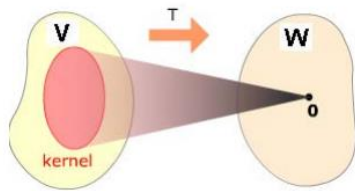
But then, since T is linear $T(u - v) = T(u) - T(v) = b - b = 0$.

The vector $u - v$ is not zero, since $u \neq v$. Hence the equation $T(x) = 0$ has more than one solution. So either the two conditions in the theorem are both true and they are both false.

Kernel of a Linear Transformation

A linear map (or transformation, or function) $f: S \rightarrow T$ transforms elements of a vector space S called domain into elements of another vector space T called codomain.

The kernel (or null space) of a linear transformation is the subset of the domain that is transformed into the zero vector.



One-One Linear Transformation

A transformation $T : R^n \rightarrow R^m$ is one-to-one if, for every vector b in R^m , the equation $T(x) = b$ has at most one solution x in R^n .

Lecture 11

Matrix Operations

Matrix Operations

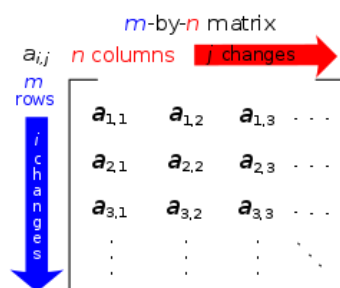
Matrix operations mainly involve three algebraic operations, which are the addition of matrices, subtraction of matrices, and multiplication of matrices. Matrix is a rectangular array of numbers or expressions arranged in rows and columns. Important applications of matrices can be found in Mathematics.

(i-j)th Element of a matrix

The element a_{ij} appearing in the i^{th} row and j^{th} column of the matrix is said to be the ij^{th} element. The matrix is also denoted by the symbol $(a_{ij})_{m,n}$.

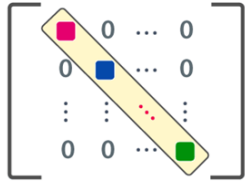
Horizontal lines of a matrix are called rows. Vertical lines of a matrix are called columns.

Each number or function a_{ij} is called its element.



Diagonal Matrix

A square matrix in which every element except the principal diagonal elements is zero is called a Diagonal Matrix. A square matrix $D = [d_{ij}]_{n \times n}$ will be called a diagonal matrix if $d_{ij} = 0$, whenever i is not equal to j .



Null Matrix or Zero Matrixes

An $m \times n$ matrix whose entries are all zero is a Null or zero matrixes and is always written as O . A null matrix may be of any order.

Equal Matrices

Two matrices are said to be equal if they have the same size (i.e., the same number of rows and columns) and same corresponding entries.

Matrix Multiplication

Matrix multiplication, also known as matrix product and the multiplication of two matrices, produces a single matrix. It is a type of binary operation. If A and B are the two matrices, then the product of the two matrices A and B are denoted by: $X = AB$.

Example:

Find the product AB for

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix}$$

Solution It follows from definition that the product AB is formed in a column-by-column manner by multiplying the successive columns of B by A . The computations are

$$\begin{matrix} c_1 & c_2 & c_3 \end{matrix} \quad \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} = 4c_1 + 0c_2 + 2c_3 = (4) \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (0) \begin{bmatrix} 2 \\ 6 \end{bmatrix} + (2) \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \end{bmatrix}$$

$$\text{Similarly,} \quad \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 7 \end{bmatrix} = (1) \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (-1) \begin{bmatrix} 2 \\ 6 \end{bmatrix} + (7) \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 27 \\ -4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix} = (4) \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (3) \begin{bmatrix} 2 \\ 6 \end{bmatrix} + (5) \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 30 \\ 26 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} = (3) \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (1) \begin{bmatrix} 2 \\ 6 \end{bmatrix} + (2) \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 13 \\ 12 \end{bmatrix}$$

Thus,
$$AB = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} = \begin{bmatrix} 12 & 27 & 30 & 13 \\ 8 & -4 & 26 & 12 \end{bmatrix}$$

Row-Column Rule for Computing AB

The number of rows of the resulting matrix equals the number of rows of the first matrix, and the number of columns of the resulting matrix equals the number of columns of the second matrix.

Example:

Use the dot product rule to compute the individual entries in the product of

AB where $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix}$.

Solution Since A has size 2×3 and B has size 3×4 , the product AB is a 2×4 matrix of the form

$$AB = \begin{bmatrix} r_1(A) \times c_1(B) & r_1(A) \times c_2(B) & r_1(A) \times c_3(B) & r_1(A) \times c_4(B) \\ r_2(A) \times c_1(B) & r_2(A) \times c_2(B) & r_2(A) \times c_3(B) & r_2(A) \times c_4(B) \end{bmatrix}$$

where $r_1(A)$ and $r_2(A)$ are the row vectors of A and $c_1(B), c_2(B), c_3(B)$ and $c_4(B)$ are the column vectors of B . For example, the entry in row 2 and column 3 of AB can be computed as

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} = \begin{bmatrix} \square & \square & \square & \square \\ \square & \square & \boxed{26} & \square \end{bmatrix}$$

$$(2 \times 4) + (6 \times 3) + (0 \times 5) = 26$$

and the entry in row 1 and column 4 of AB can be computed as

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} = \begin{bmatrix} \square & \square & \square & \boxed{13} \\ \square & \square & \square & \square \end{bmatrix}$$

$$(1 \times 3) + (2 \times 1) + (4 \times 2) = 13$$

Here is the complete set of computations:

$$\begin{aligned}
 (AB)_{11} &= (1 \times 4) + (2 \times 0) + (4 \times 2) = 12 \\
 (AB)_{12} &= (1 \times 1) + (2 \times -1) + (4 \times 7) = 27 \\
 (AB)_{13} &= (1 \times 4) + (2 \times 3) + (4 \times 5) = 30 \\
 (AB)_{14} &= (1 \times 3) + (2 \times 1) + (4 \times 2) = 13 \\
 (AB)_{21} &= (2 \times 4) + (6 \times 0) + (0 \times 2) = 8 \\
 (AB)_{22} &= (2 \times 1) + (6 \times -1) + (0 \times 7) = -4 \\
 (AB)_{23} &= (2 \times 4) + (6 \times 3) + (0 \times 5) = 26 \\
 (AB)_{24} &= (2 \times 3) + (6 \times 1) + (0 \times 2) = 12
 \end{aligned}$$

Properties of Matrix Multiplication

- a. $A(BC) = (AB)C$ (associative law of multiplication)
- b. $A(B + C) = AB + AC$ (left distributive law)
- c. $(B + C)A = BA + CA$ (right distributive law)
- d. $r(AB) = (rA)B = A(rB)$ (for any scalar r)
- e. $I_m A = A = A I_m$ (identity for matrix multiplication)

Transpose of a Matrix

The transpose of a matrix is found by interchanging its rows into columns or columns into rows. The transpose of the matrix is denoted by using the letter “T” in the superscript of the given matrix. For example, if “A” is the given matrix, then the transpose of the matrix is represented by A' or A^T .

Lecture 12

The Inverse of a Matrix

Inverse of a square Matrix

If A is an $n \times n$ matrix, a matrix C of order $n \times n$ is called multiplicative inverse of A if $AC = CA = I$ where I is the $n \times n$ identity matrix.

Invertible Matrix

If the inverse of a square matrix exists, it is called an invertible matrix. In this case, we say that A is invertible and we call C an inverse of A.

Singular matrix and non-singular matrix

A matrix that is not invertible is sometimes called a singular matrix, and an invertible matrix is called a non-singular matrix.

- a. If A is an invertible matrix, then A^{-1} is invertible and $(A^{-1})^{-1} = A$
- b. If A and B are $n \times n$ invertible matrices, then so is AB , and the inverse of AB is the product of the inverses of A and B in the reverse order. That is $(AB)^{-1} = B^{-1}A^{-1}$
- c. If A is an invertible matrix, then so is A^T , and the inverse of A^T is the transpose of A^{-1} . That is $(A^T)^{-1} = (A^{-1})^T$

Elementary Matrices

There are three types of elementary row operations that can be performed on a matrix:
There are three types of elementary operations

1. Interchanging of any two rows
2. Multiplication to a row by a nonzero constant
3. Adding a multiple of one row to another

Elementary matrix

An elementary matrix is a square matrix that has been obtained by performing an elementary row or column operation on an identity matrix.

The operations on the right side of this table are called the inverse operations of the corresponding operations on the left side.

Row operation on I that produces E	Row operation on E that reproduces I
Multiply row i by $c \neq 0$	Multiply row i by $1/c$
Interchange rows i and j	Interchange rows i and j
Add c times row i to row j	Add $-c$ times row i to row j

Example

Find the inverse of the given matrix if possible $A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$

Solution $\det A = \begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{vmatrix} = -1$

As the given matrix is non-singular therefore, inverse is possible.

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -2 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

$R_2 - 2R_1, R_3 - 3R_1$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}$$

$R_3 - R_2$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 1 & -1 \end{bmatrix}$$

Multiply R_3 by -1

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & 1 & -1 \end{bmatrix}$$

$R_1 - R_3, R_2 + R_3$

Hence the inverse of matrix A is $A^{-1} = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & 1 & -1 \end{bmatrix}$

Lecture 13

Characterizations of Invertible Matrices

Theorem

Let A be a For any b For any $b \in \mathbb{R}^n$, the equation $Ax = b$ (1) Has the unique solution i.e. $x = A^{-1} * b$.

Proof

Since A is invertible and $b \in \mathbb{R}^n$, be any vector. Then, we must have a matrix $A^{-1}b$ which is a solution of eq. (1) .i.e. $Ax = A(A^{-1}b) = Ib = b$.

Uniqueness

For uniqueness, we assume that there is another solution u . Indeed, it is a solution of eq. (1) so it must be $u = A^{-1}b$, it means $x = A^{-1}b = u$. This shows that $u = x$.

Theorem

Let A and B be the square matrices such that $AB = I$. Then, A and B are invertible with $B = A^{-1}$ and $A = B^{-1}$

Example 1

Solve the system of linear equations

$$\begin{aligned} 2x_1 + x_2 + x_3 &= 1 \\ 5x_1 + x_2 + 3x_3 &= 3 \\ x_1 + 4x_3 &= 6 \end{aligned}$$

Solution

Consider the linear system

$$\begin{aligned} 2x_1 + x_2 + x_3 &= 1 \\ 5x_1 + x_2 + 3x_3 &= 3 \\ x_1 + 4x_3 &= 6 \end{aligned}$$

The Matrix form of system is $Ax = b$, where

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 5 & 1 & 3 \\ 1 & 0 & 4 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix}$$

Here, $\det(A) = 2(4) - 1(20 - 3) + 1(0 - 1) = 8 - 17 - 1 = -10 \neq 0$

So, A is invertible. Now, we apply the inversion algorithm:

$$\begin{aligned} &\left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 5 & 1 & 3 & 0 & 1 & 0 \\ 1 & 0 & 4 & 0 & 0 & 1 \end{array} \right] \\ &\left[\begin{array}{ccc|ccc} 1 & 0 & 4 & 0 & 0 & 1 \\ 5 & 1 & 3 & 0 & 1 & 0 \\ 2 & 1 & 1 & 1 & 0 & 0 \end{array} \right] \\ &\left[\begin{array}{ccc|ccc} 1 & 0 & 4 & 0 & 0 & 1 \\ 0 & 1 & -17 & 0 & 1 & -5 \\ 0 & 1 & -7 & 1 & 0 & -2 \end{array} \right], \quad -5R_1 + R_2, \quad -2R_1 + R_3 \\ &\left[\begin{array}{ccc|ccc} 1 & 0 & 4 & 0 & 0 & 1 \\ 0 & 1 & -17 & 0 & 1 & -5 \\ 0 & 0 & 10 & 1 & -1 & 3 \end{array} \right], \quad -R_2 + R_3, \end{aligned}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 4 & 0 & 0 & 1 \\ 0 & 1 & -17 & 0 & 1 & -5 \\ 0 & 0 & 1 & \frac{1}{10} & \frac{-1}{10} & \frac{3}{10} \end{array} \right], \frac{R_3}{10}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{-2}{5} & \frac{2}{5} & \frac{-1}{5} \\ 0 & 1 & 0 & \frac{17}{10} & \frac{-7}{10} & \frac{1}{10} \\ 0 & 0 & 1 & \frac{1}{10} & \frac{-1}{10} & \frac{3}{10} \end{array} \right], \begin{array}{l} -4R_3 + R_1 \\ -17R_3 + R_2 \end{array}$$

Hence, $A^{-1} = \begin{bmatrix} \frac{-2}{5} & \frac{2}{5} & \frac{-1}{5} \\ \frac{17}{10} & \frac{-7}{10} & \frac{1}{10} \\ \frac{1}{10} & \frac{-1}{10} & \frac{3}{10} \end{bmatrix}$

Thus, the solution of the linear system is $x = A^{-1}b = A^{-1} \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} \frac{-2}{5} & \frac{2}{5} & \frac{-1}{5} \\ \frac{17}{10} & \frac{-7}{10} & \frac{1}{10} \\ \frac{1}{10} & \frac{-1}{10} & \frac{3}{10} \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} \frac{-2}{5} \\ \frac{1}{5} \\ \frac{8}{5} \end{bmatrix}$

Example 3

Solve the system of linear equation

$$x_1 + 6x_2 + 4x_3 = 2$$

$$2x_1 + 4x_2 - x_3 = 3$$

$$-x_1 + 2x_2 + 5x_3 = 3$$

Solution

The matrix coefficient

$$A = \begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{bmatrix}$$

$$\det(A) = 1(20 + 2) - 6(10 - 1) + 4(4 + 4)$$

$$= 22 - 54 + 32$$

$$= 0$$

Thus, A is not invertible. Hence, the inversion method fails.

Theorem (Invertible Matrix Theorem)

Let A be a square $n \times n$ matrix. Then the following statements are equivalent. (Means if any one holds then all are true).

- (a) A is an invertible matrix.
- (b) A is row equivalent to the $n \times n$ identity matrix.
- (c) A has n pivot positions.
- (d) The equation $Ax = 0$ has only the trivial solution.
- (e) The columns of A form a linearly independent set.
- (f) The linear transformation $x \rightarrow Ax$ is one-to-one.
- (g) The equation $Ax = b$ has at least one solution for each b in $\mathbb{R}^n$.
- (h) The columns of A span $\mathbb{R}^n$.
- (i) The linear transformation $x \rightarrow Ax$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
- (j) There is a $n \times n$ matrix C such that $CA = I$.
- (k) There is a $n \times n$ matrix D such that $AD = I$.
- (l) A^T is an invertible matrix.

Example 6

Show that the matrix $A = \begin{bmatrix} 1 & 0 & -4 \\ 1 & 1 & 5 \\ 0 & 1 & 2 \end{bmatrix}$ is invertible by using Invertible Matrix Theorem

Solution

By row equivalent,

$$A = \begin{bmatrix} 1 & 0 & -4 \\ 1 & 1 & 5 \\ 0 & 1 & 2 \end{bmatrix}, -R_1 + R_2$$

$$\begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 9 \\ 0 & 1 & 2 \end{bmatrix}, -R_2 + R_3$$

$$\begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 9 \\ 0 & 0 & -7 \end{bmatrix}$$

It shows that A has three pivot positions and hence is invertible, by the Invertible Matrix Theorem (c).

Example 7

Use the Invertible Matrix Theorem to decide if A is invertible, where

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & -2 \\ -5 & -1 & 9 \end{bmatrix}$$

Solution

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{bmatrix} \quad \text{Here, } A \text{ has three pivot positions and hence is}$$

invertible by the Invertible Matrix Theorem (c).

Invertible Linear Transformations

Recall that matrix multiplication corresponds to composition of linear transformations. When a matrix A is invertible, the equation $A^{-1}Ax = x$ can be viewed as a statement about linear transformations.

Lecture 14

Partitioned Matrices

Partitioned Matrices

A block matrix or a partitioned matrix is a partition of a matrix into rectangular smaller matrices called blocks.

General Partitioning of a Matrix

A matrix can be partitioned (subdivided) into sub matrices (also called blocks) in various ways by inserting lines between selected rows and columns.

Example

Let A be a general matrix of 5×3 order, we have

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \\ a_{51} & a_{52} & a_{53} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

Partition (a)

$$A = \left[\begin{array}{c|c|c} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \\ a_{51} & a_{52} & a_{53} \end{array} \right] = [c_1 \mid c_2 \mid c_3]$$

Partition (b)

The c_i in the partitioned matrix are sometimes called the column matrices of A.

$$A = \left[\begin{array}{c|c|c} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \\ a_{51} & a_{52} & a_{53} \end{array} \right] = \left[\begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array} \right]$$

Partition (c)

The r_i in the partitioned matrix are sometimes called the row matrices of A.

Toeplitz matrix

A matrix in which each descending diagonal from left to right is constant is called a Toeplitz matrix or diagonal-constant matrix.

Block Toeplitz matrix

A blocked matrix in which blocks (blocked matrices) are repeated down the diagonals of the matrix is called a blocked Toeplitz matrix.

A block Toeplitz matrix **B** has the form

$$B = \begin{bmatrix} B(1,1) & B(1,2) & B(1,3) & B(1,4) & B(1,5) \\ B(2,1) & B(1,1) & B(1,2) & B(1,3) & B(1,4) \\ B(3,1) & B(2,1) & B(1,1) & B(1,2) & B(1,3) \\ B(4,1) & B(3,1) & B(2,1) & B(1,1) & B(1,2) \\ B(5,1) & B(4,1) & B(3,1) & B(2,1) & B(1,1) \end{bmatrix}$$

Block Diagonal Matrices

A partitioned matrix A is said to be block diagonal if the matrices on the main diagonal are square and all other position matrices are zero, i.e.

$$A = \begin{bmatrix} D_1 & 0 & \dots & 0 \\ 0 & D_2 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & D_k \end{bmatrix} \quad (1)$$

where the matrices $D_1, D_2, \dots, D_k$ are square. It can be shown that the matrix A in (1) is invertible if and only if each matrix on the diagonal is invertible. i.e.

$$A^{-1} = \begin{bmatrix} D_1^{-1} & 0 & \dots & 0 \\ 0 & D_2^{-1} & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & D_k^{-1} \end{bmatrix}$$

Block Upper Triangular Matrices

A partitioned square matrix A is said to be block upper triangular if the matrices on the main diagonal are square and all matrices below the main diagonal are zero; that is, the matrix is partitioned as

$$A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1k} \\ 0 & A_{22} & \dots & A_{2k} \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & A_{kk} \end{bmatrix} \text{ where the matrices } A_{11}, A_{22}, \dots, A_{kk} \text{ are}$$

square.

Example

Find A^{-1} of

$$A = \begin{bmatrix} 1 & 9 & -5 & 0 \\ 3 & 3 & 3 & -2 \\ 0 & 0 & 7 & 0 \\ 0 & 0 & 3 & 1 \end{bmatrix}$$

Solution

Let partition given matrix A in form

$$A = \left[\begin{array}{cc|cc} 1 & 9 & -5 & 0 \\ 3 & 3 & 3 & -2 \\ \hline 0 & 0 & 7 & 0 \\ 0 & 0 & 3 & 1 \end{array} \right]$$

$$\text{Put } A_{11} = \begin{bmatrix} 1 & 9 \\ 3 & 3 \end{bmatrix}, A_{12} = \begin{bmatrix} -5 & 0 \\ 3 & -2 \end{bmatrix} \text{ and } A_{22} = \begin{bmatrix} 7 & 0 \\ 3 & 1 \end{bmatrix}$$

$$\text{Thus } A_{11}^{-1} = \frac{\text{Adj}A}{\det(A)} = \frac{\begin{bmatrix} 3 & -9 \\ -3 & 1 \end{bmatrix}}{-24} = \begin{pmatrix} \frac{1}{8} & \frac{9}{24} \\ -\frac{3}{8} & \frac{1}{-24} \end{pmatrix} \text{ and } A_{22}^{-1} = \begin{bmatrix} \frac{1}{7} & 0 \\ -\frac{3}{7} & 1 \end{bmatrix}$$

Moreover,

$$-A_{11}^{-1}A_{12}A_{22}^{-1} = \begin{pmatrix} \frac{1}{-8} & \frac{9}{24} \\ \frac{1}{8} & \frac{1}{-24} \end{pmatrix} \begin{bmatrix} -5 & 0 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} \frac{1}{7} & 0 \\ -\frac{3}{7} & 1 \end{bmatrix} = \begin{pmatrix} \frac{-4}{7} & \frac{3}{4} \\ \frac{1}{4} & \frac{-1}{12} \end{pmatrix}$$

So by (5), we have

$$A^{-1} = \begin{bmatrix} \frac{-1}{8} & \frac{3}{8} & \frac{-4}{7} & \frac{3}{4} \\ \frac{1}{8} & \frac{-1}{24} & \frac{1}{4} & \frac{-1}{12} \\ 0 & 0 & \frac{1}{7} & 0 \\ 0 & 0 & \frac{-3}{7} & 1 \end{bmatrix}$$

Lecture 15 Matrix Factorizations

Matrix Factorization

A factorization of a matrix as a product of two or more matrices is called Matrix Factorization.

Uses of Matrix Factorization

Matrix factorizations will appear at a number of key points throughout the course. This lecture focuses on a factorization that lies at the heart of several important computer programs widely used in applications.

LU Factorization or LU-decomposition

LU factorization is another name as LU decomposition, as the both titles indicate that a given matrix can be expressed in two smaller matrices, which include an upper triangular matrix and a lower triangular matrix. The product of these two matrices reveals the given matrix

LU Decomposition with Pivoting

$$\begin{array}{rrrrr} x_1 & & +2x_3 & +3x_4 & =1 \\ -x_1 & +2x_2 & +2x_3 & -3x_4 & =-1 \\ & x_2 & +x_3 & +4x_4 & =2 \\ 6x_1 & +2x_2 & +2x_3 & +4x_4 & =1 \end{array} \quad [A \vdots b] = \left[\begin{array}{cccc|c} 1 & 0 & 2 & 3 & 1 \\ -1 & 2 & 2 & -3 & -1 \\ 0 & 1 & 1 & 4 & 2 \\ 6 & 2 & 2 & 4 & 1 \end{array} \right]$$

Start with

$$[A] = \begin{bmatrix} 1 & 0 & 2 & 3 \\ -1 & 2 & 2 & -3 \\ 0 & 1 & 1 & 4 \\ 6 & 2 & 2 & 4 \end{bmatrix} \quad [L] = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \quad [P] = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

No need to consider $\{b\}$ in decomposition

If a square matrix A can be reduced to row echelon form with no row interchanges, then A has an LU-decomposition.

Gaussian elimination

Gaussian elimination is a method in which an augmented matrix is subjected to row operations until the component corresponding to the coefficient matrix is reduced to triangular form.

After we have obtained our triangular matrix, there are two different approaches we can use to solve a system of linear equations:

1. Forward substitution
2. Back substitution

$$\begin{bmatrix} a_{11} & 0 & 0 & \dots & 0 \\ a_{21} & a_{22} & 0 & \dots & 0 \\ a_{31} & a_{32} & a_{33} & \dots & 0 \\ \dots & \dots & \dots & \dots & 0 \\ a_{n1} & a_{n2} & \dots & a_{n,n-1} & a_{n,n} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix}$$

Forward substitution

the procedure of solving a system of linear algebraic equations (SLAE) with a lower triangular coefficient matrix is known as forward substitution. Solving an SLAE with a triangular matrix form is a variant of the generic substitution approach.

Equation

$$Lx=y$$

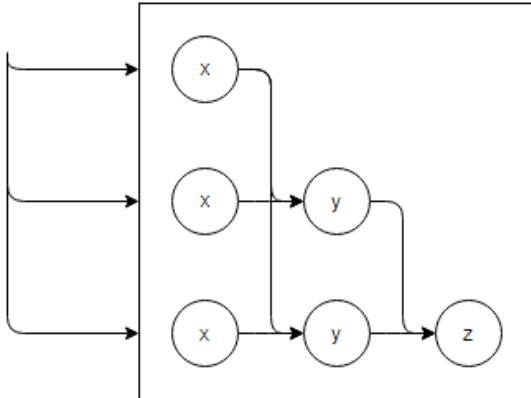
- L represents the factor of the matrix of the lower triangle.
- x represents the variable of matrix.
- y is the result vector.

The matrix form of a lower triangle:

$$\begin{bmatrix} x & 0 & 0 \\ 2x & y & 0 \\ 3x & 2y & 3z \end{bmatrix}$$

Visualization of forward substitution

The visualization shows how forward substitution works. The method transforms the matrix into a lower triangular form and then starts solving an equation from top to bottom.



- The diagram above shows how forward substitution works. In this process, we make a lower triangle and start from the top.
- As we can see at the top, only x exists, and other values are zero, so it is easy to find a value of x and use it for the next step.
- In the second step, we find the value of y by using the value of x , which came from the first step.
- Similarly, in the third step, we use x and y values and find the value of z .

Back substitution

The procedure of solving an SLAE with an upper triangular coefficient matrix is known as **back substitution**.

Equation

$$Ux=y$$

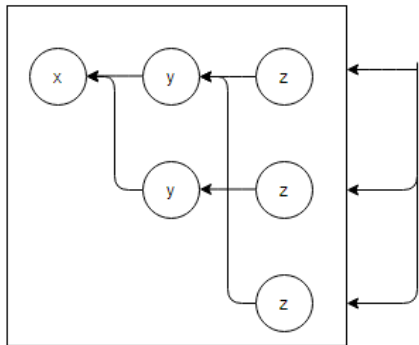
- U represents the factor of matrix of upper triangle.
- x represents the variable of matrix.
- y is the result vector.

The matrix form of an upper triangle:

$$\begin{bmatrix} 3x & 2y & 3z \\ 0 & y & 2z \\ 0 & 0 & z \end{bmatrix}$$

Visualization of backward substitution

It shows how the backward substitution works. The method transforms the matrix into an upper triangular form and then starts solving an equation from bottom to top.



- The lower diagram shows how back substitution works. In this process we make an upper triangle and start from the bottom.
- As we can see at the bottom, only z exists and other values are zero, so it is easy to find a value of z and use it for next step.
- In the second step, we find the value of y using the value of z, which came from previous step.
- Similarly, in the third step, we use y and z to find the value of x.

Lecture 16

Iterative Solutions of Linear Systems

Iterative Methods for Linear Systems

One of the most important and common applications of numerical linear algebra is the solution of linear systems that can be expressed in the form $A*x = b$. When A is a large sparse matrix, you can solve the linear system using iterative methods, which enable you to trade-off between the run time of the calculation and the precision of the solution. This topic describes the iterative methods available in MATLAB to solve the equation $A*x = b$.

In computational mathematics, an **iterative method** is a mathematical procedure that uses an initial value to generate a sequence of improving approximate solutions for a class of problems, in which the n -th approximation is derived from the previous ones.

A specific implementation with termination criteria for a given iterative method like gradient descent, hill climbing, Newton's method, or quasi-Newton methods like BFGS, is an algorithm of the iterative method. An iterative method is called **convergent** if the corresponding sequence converges for given initial approximations. A mathematically rigorous convergence analysis of an iterative method is usually performed; however, heuristic-based iterative methods are also common.

In contrast, **direct methods** attempt to solve the problem by a finite sequence of operations. In the absence of rounding errors, direct methods would deliver an exact solution (for example, solving a linear system of equations $Ax=b$ by Gaussian elimination). Iterative methods are often the only choice for nonlinear equations. However, iterative methods are often useful even for linear problems involving many variables (sometimes on the order of millions), where direct methods would be prohibitively expensive (and in some cases impossible) even with the best available computing power.

If an equation can be put into the form $f(x) = x$, and a solution x is an attractive fixed point of the function f , then one may begin with a point x_1 in the basin of attraction of x , and let $x_{n+1} = f(x_n)$ for $n \geq 1$, and the sequence $\{x_n\}_{n \geq 1}$ will converge to the solution x . Here x_n is the n th approximation or iteration of x and x_{n+1} is the next or $n + 1$ iteration of x . Alternately, superscripts in parentheses are often used in numerical methods, so as not to interfere with subscripts with other meanings. (For example, $x^{(n+1)} = f(x^{(n)})$.) If the function f is continuously differentiable, a sufficient condition for convergence is that the spectral radius of the derivative is strictly bounded by one in a neighborhood of the fixed point. If this condition holds at the fixed point, then a sufficiently small neighborhood (basin of attraction) must exist.

The Gauss-Seidel Method

In numerical linear algebra, the **Gauss–Seidel method**, also known as the **Liebmann method** or the **method of successive displacement**, is an iterative method used to solve a system of linear equations. It is named after the German mathematicians Carl Friedrich Gauss and Philipp Ludwig von Seidel, and is similar to the Jacobi method.

The Gauss–Seidel method is an iterative technique for solving a square system of n linear

equations. Let $Ax = b$ be a square system of n linear equations, where:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

Alternative Approach

Alternative theories are number oriented rather than digit oriented, calculated from the left rather than the right, and are flexible options as opposed to the ‘one right way’. They have many benefits which include students have a greater understanding of numbers and as a result there are fewer errors and less reteaching required.

Comparison of Jacobi’s and Gauss-Seidel method

The difference between the Gauss–Seidel and Jacobi methods is that the Jacobi method uses the values obtained from the previous step while the Gauss–Seidel method always applies the latest updated values during the iterative procedures, as demonstrated.

Condition for the Convergence of both Iterative Methods

If the function f is continuously differentiable, a sufficient condition for convergence is that the spectral radius of the derivative is strictly bounded by one in a neighborhood of the fixed point. If this condition holds at the fixed point, then a sufficiently small neighborhood (basin of attraction) must exist.

An $n \times n$ matrix A is said to be strictly diagonally dominant if the absolute value of each diagonal entry exceeds the sum of the absolute values of the other entries in the same row.

Lecture 17

Introduction to Determinant

In algebra, the determinant is a special number associated with any square matrix. As we have studied in earlier classes, that the determinant of 2×2 matrix is defined as the product of the entries on the main diagonal minus the product of the entries off the main diagonal. The determinant of a matrix A is denoted by $\det(A)$ or $|A|$.

For example:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then
or

$$\det(A) = ad - bc.$$

$$|A| = ad - bc$$

Example

Find the determinant of the matrix $A = \begin{bmatrix} 1 & 4 & 3 \\ 5 & 2 & 4 \\ 3 & 6 & 3 \end{bmatrix}$

Solution Given $A = \begin{bmatrix} 1 & 4 & 3 \\ 5 & 2 & 4 \\ 3 & 6 & 3 \end{bmatrix}$

$$\begin{aligned} |A| &= \begin{vmatrix} 1 & 4 & 3 \\ 5 & 2 & 4 \\ 3 & 6 & 3 \end{vmatrix} \\ &= 1 \begin{vmatrix} 2 & 4 \\ 6 & 3 \end{vmatrix} - 4 \begin{vmatrix} 5 & 4 \\ 3 & 3 \end{vmatrix} + 3 \begin{vmatrix} 5 & 2 \\ 3 & 6 \end{vmatrix} \\ &= 1(2 \times 3 - 4 \times 6) - 4(5 \times 3 - 4 \times 3) + 3(5 \times 6 - 2 \times 3) \\ &= 1(6 - 24) - 4(15 - 12) + 3(30 - 6) \\ &= 1(-18) - 4(3) + 3(24) \\ &= -18 - 12 + 72 \\ &= 42 \end{aligned}$$

Cofactor of an element

A cofactor is a number that is obtained by eliminating the row and column of a particular element which is in the form of a square or rectangle. The cofactor is preceded by a negative or positive sign based on the element's position.

Determinants of Triangular Matrices

Determinants of the triangular matrices are also easy to evaluate regardless of size.

Example

$$\begin{vmatrix} -2 & 5 & 7 \\ 0 & 3 & 8 \\ 0 & 0 & 5 \end{vmatrix} = (-2)(3)(5) = -30$$

$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 4 & 9 & 0 & 0 \\ -7 & 6 & -1 & 0 \\ 3 & 8 & -5 & -2 \end{vmatrix} = (1)(9)(-1)(-2) = 18$$

$$\begin{vmatrix} 1 & 2 & 7 & -3 \\ 0 & 1 & -4 & 1 \\ 0 & 0 & 2 & 7 \\ 0 & 0 & 0 & 3 \end{vmatrix} = (1)(1)(2)(3) = 6$$

Lecture 18

Properties of Determinants

Some of them have already been discussed and you will be familiar with these. These properties become helpful, while computing the values of the determinants. The secret of determinants lies in how they change when row or column operations are performed.

(Row Operations):

Let A be a square matrix.

- If a multiple of one row of A is added to another row, the resulting determinant will remain same.
- If two rows of A are interchanged to produce B , then $\det B = -\det A$.
- If one row of A is multiplied by k to produce B , then $\det B = k \cdot \det A$.

Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 1 & 1 \\ 0 & 8 & 9 \end{bmatrix}$$

$$\text{Now, } \det A = \begin{vmatrix} 1 & 2 & 3 \\ 5 & 1 & 1 \\ 0 & 8 & 9 \end{vmatrix} = 1(9-8) - 2(45-0) + 3(40-0) = 1 - 90 + 120 = 31$$

$$\text{Now interchange column 1st with 2}^{\text{nd}} \text{ we get a new matrix, } B = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 5 & 1 \\ 8 & 0 & 9 \end{bmatrix}$$

$$\det B = \begin{vmatrix} 2 & 1 & 3 \\ 1 & 5 & 1 \\ 8 & 0 & 9 \end{vmatrix} = 2(45-0) - 1(9-8) + 3(0-40) = 90 - 1 - 120 = -31$$

- c. If one row of A is multiplied by k to produce B , then $\det B = k \cdot \det A$.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 0 & 1 \\ 0 & 8 & 9 \end{bmatrix}$$

$$|A| = 1(0-8) - 2(45-0) + 3(40-0) \\ = -8 - 90 + 120 = 22$$

Multiplying R_1 by k , we get say

$$B = \begin{bmatrix} 1k & 2k & 3k \\ 5 & 0 & 1 \\ 0 & 8 & 9 \end{bmatrix}$$

$$\begin{aligned}
 |B| &= k(40 - 0) - 2k(45 - 0) + 3k(40 - 0) \\
 &= 40k - 90k + 120k = 22k \\
 &= k|A|
 \end{aligned}$$

An Algorithm to evaluate the determinant

Algorithm means a sequence of a finite number of steps to get a desired result. The word Algorithm comes from the famous Muslim mathematician AL-Khwarizmi who invented the word algebra.

The step-by-step evaluation of $\det(A)$ of order n is obtained as follows:

- Step 1: By an interchange of rows of A (and taking the resulting sign into account) bring a non zero entry to $(1,1)$ the position (unless all the entries in the first column are zero in which case $\det A=0$).
- Step 2: By adding suitable multiples of the first row to all the other rows, reduce the $(n-1)$ entries, except $(1,1)$ in the first column, to 0. Expand $\det(A)$ by its first column. Repeat this process or continue the following steps.
- Step 3: Repeat step 1 and step 2 with the last remaining rows concentrating on the second column.
- Step 4: Repeat step 1, step 2 and step 3 with the remaining $(n-2)$ rows, $(n-3)$ rows and so on, until a triangular matrix is obtained.
- Step 5: Multiply all the diagonal entries of the resulting triangular matrix and then multiply it by its sign to get $\det(A)$.

Lecture 19

Cramer's Rule, Volume, and Linear Transformations

Cramer's Rule

Cramer's rule is needed in a variety of theoretical calculations. For instance, it can be used to study how the solution of $Ax = b$ is affected by changes in the entries of b . However, the formula is inefficient for hand calculations, except for 2×2 or perhaps 3×3 matrices.

Theorem 1 (Cramer's Rule) Let A be an invertible $n \times n$ matrix. For any b in R^n , the unique solution x of $Ax = b$ has entries given by

$$x_i = \frac{\det A_i(b)}{\det A}, \quad i = 1, 2, \dots, n$$

Example 1 Use Cramer's rule to solve the system

$$3x_1 - 2x_2 = 6$$

$$-5x_1 + 4x_2 = 8$$

Solution Write the system in matrix form, $Ax = b$

$$\begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

where

$$A = \begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \& b = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

$$\det A = \begin{vmatrix} 3 & -2 \\ -5 & 4 \end{vmatrix} = 12 - 10 = 2$$

$$A_1(b) = \begin{bmatrix} 6 & -2 \\ 8 & 4 \end{bmatrix}, A_2(b) = \begin{bmatrix} 3 & 6 \\ -5 & 8 \end{bmatrix}$$

Since $\det A = 2$, the system has a unique solution. By Cramer's rule,

$$x_1 = \frac{\det A_1(b)}{\det A} = \frac{24 + 16}{2} = 20$$

$$x_2 = \frac{\det A_2(b)}{\det A} = \frac{24 + 30}{2} = 27$$

Solve, by Cramer's Rule, the system of equations

$$2x_1 - x_2 + 3x_3 = 1$$

$$x_1 + 2x_2 - x_3 = 2$$

$$3x_1 + 2x_2 + 2x_3 = 3$$

Solution Here $A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & -1 \\ 3 & 2 & 2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, A_1 = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 2 & -1 \\ 3 & 2 & 2 \end{bmatrix}$

$$A_2 = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & -1 \\ 3 & 3 & 2 \end{bmatrix}, A_3 = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 2 \\ 3 & 2 & 3 \end{bmatrix}$$

$$D = \det A = 2 \cdot 6 - 1 \cdot (-5) + 3(-4) = 5$$

$$D_1 = \det A_1 b = 1 \cdot 6 + 2 \cdot 8 + 3(-5) = 7$$

$$D_2 = \det A_2 b = 2 \cdot (7) - 1 \cdot (5) + 3(-3) = 0$$

$$D_3 = \det A_3 b = 2 \cdot (2) + 1 \cdot (-3) + 1(-4) = -3$$

$$\text{So } x_1 = \frac{D_1}{D} = \frac{7}{5}, \quad x_2 = \frac{D_2}{D} = 0, \quad x_3 = \frac{D_3}{D} = -\frac{3}{5}$$

Example 4 Use Cramer's Rule to solve.

$$\begin{aligned}x_1 + 2x_3 &= 6 \\-3x_1 + 4x_2 + 6x_3 &= 30 \\-x_1 - 2x_2 + 3x_3 &= 8\end{aligned}$$

Solution

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -3 & 4 & 6 \\ -1 & -2 & 3 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 30 \\ 8 \end{bmatrix}$$

$$\therefore A_1 = \begin{bmatrix} 6 & 0 & 2 \\ 30 & 4 & 6 \\ 8 & -2 & 3 \end{bmatrix}, A_2 = \begin{bmatrix} 1 & 6 & 2 \\ -3 & 30 & 6 \\ -1 & -8 & 3 \end{bmatrix}, A_3 = \begin{bmatrix} 1 & 0 & 6 \\ -3 & 4 & 30 \\ -1 & -2 & 8 \end{bmatrix}$$

Therefore,

$$\begin{aligned}x_1 &= \frac{\det(A_1b)}{\det(A)} = \frac{-40}{44} = \frac{-10}{11}, & x_2 &= \frac{\det(A_2b)}{\det(A)} = \frac{72}{44} = \frac{18}{11} \\x_3 &= \frac{\det(A_3b)}{\det(A)} = \frac{152}{44} = \frac{38}{11}\end{aligned}$$

Inverse of Matrix

The **inverse of matrix** is a matrix, which on multiplication with the given matrix gives the multiplicative identity. For a square matrix A , its inverse is A^{-1} , and $A \cdot A^{-1} = A^{-1} \cdot A = I$, where I is the identity matrix. The matrix whose determinant is non-zero and for which the inverse matrix can be calculated is called an invertible matrix.

In the case of real numbers, the inverse of any real number a was the number a^{-1} , such that a times a^{-1} equals 1. We knew that for a real number, the inverse of the number was the reciprocal of the number, as long as the number wasn't zero. The inverse of a square matrix A , denoted by A^{-1} , is the matrix so that the product of A and A^{-1} is the identity matrix. The identity matrix that results will be the same size as matrix A .

The formula to find the inverse of a matrix is: $A^{-1} = 1/|A| \cdot \text{Adj } A$, where

- $|A|$ is the determinant of A and
- $\text{Adj } A$ is the adjoint of A

Inverse of a Matrix



$$A^{-1} = \frac{1}{|A|} \cdot \text{Adj } A$$

Determinants as Area or Volume

Determinants can be interpreted geometrically as areas and volumes. This might make intuitive sense if we observe that the area of a parallelogram is determined by $\mathbf{u}$ and $\mathbf{v}$. We are used to working with column vectors. In this section, however, when we use vectors to form a matrix we will regard them as row vectors.

Lecture 20

Vector Spaces and Subspaces

Vector Spaces and Subspaces

A **subspace** is a vector space that is contained within another vector space. So every subspace is a vector space in its own right, but it is also defined relative to some other (larger) vector space. We will discover shortly that we are already familiar with a wide variety of subspaces from previous sections.

S: Subspace

Suppose that V and W are two vector spaces that have identical definitions of vector addition and scalar multiplication, and that W is a subset of V , $W \subseteq V$. Then W is a *subspace* of V .

Space and W is a subset of V , $W \subseteq V$. Endow W with the same operations as V . Then W is a subspace if and only if three conditions are met

1. W is nonempty, $W \neq \emptyset$.
2. If $x \in W$ and $y \in W$, then $x + y \in W$.
3. If $\alpha \in \mathbb{C}$ and $x \in W$, then $\alpha x \in W$.

So just three conditions, plus being a subset of a known vector space, gets us all ten properties. Fabulous! This theorem can be paraphrased by saying that a subspace is “a nonempty subset (of a vector space) that is closed under vector addition and scalar multiplication.”

Axioms of Vector Space

1. Closure Property For any two vectors u & $v \in V$, implies $u + v \in V$
2. Commutative Property For any two vectors u & $v \in V$, implies $u + v = v + u$
3. Associative Property For any three vectors $u, v, w \in V$, $u + (v + w) = (u + v) + w$

4. Additive Identity For any vector $u \in V$, there exist a zero vector 0 such that $0 + u = u + 0 = u$
5. Additive Inverse For each vector $u \in V$, there exist a vector $-u$ in V such that $-u + u = 0 = u + (-u)$
6. Scalar Multiplication For any scalar k and a vector $u \in V$ implies $k u \in V$
7. Distributive Law For any scalar k if u & $v \in V$, then $k(u + v) = k u + k v$
8. For scalars m, n and for any vector $u \in V$, $(m + n) u = m u + n u$
9. For scalars m, n and for any vector $u \in V$, $m(n u) = (m n) u = n(m u)$
10. For any vector $u \in V$, $1u = u$ where 1 is the multiplicative identity of real numbers.

Lecture 21

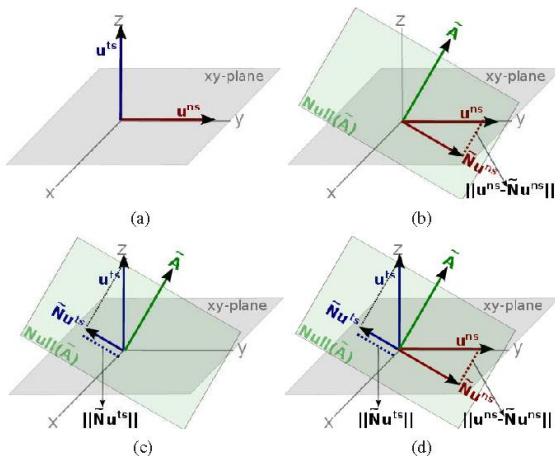
Null Spaces, Column Spaces, and Linear Transformations

Null Spaces

"Null spaces" refer to a concept in linear algebra. In the context of a matrix, the null space, also known as the kernel, represents the set of all vectors that, when multiplied by the matrix, yield the zero vector.

To put it simply, the null space of a matrix consists of all the possible solutions to the equation $Ax = 0$, where A is the matrix and x is a vector. These solutions form a subspace of the vector space in which the matrix operates.

The null space is important in various applications, such as solving systems of linear equations, understanding the rank of a matrix, and finding a basis for the solution space. It helps us explore the linear dependencies and determine the solutions to homogeneous systems of equations.

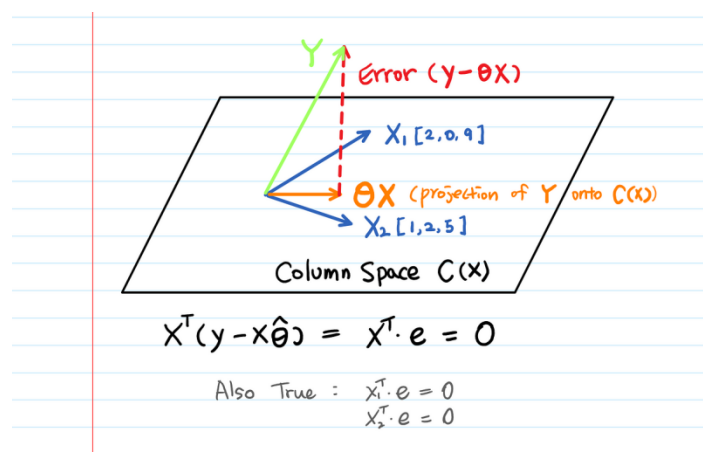


Visualization of the objective functions. (a) Example data exist with

Column Spaces

To put it simply, if you have a matrix A , the column space is the set of all possible linear combinations of the columns of A . It represents the entire range of values that can be obtained by multiplying the matrix with a vector.

The column space is useful in understanding the properties and behavior of a matrix. It helps determine the dimension of the range of a matrix, which is equivalent to the number of linearly independent columns. The column space also plays a crucial role in applications such as solving systems of linear equations, finding bases for vector spaces, and performing matrix transformations.



Linear Transformations

Linear map or linear function is a function between two vector spaces that preserves addition and scalar multiplication.

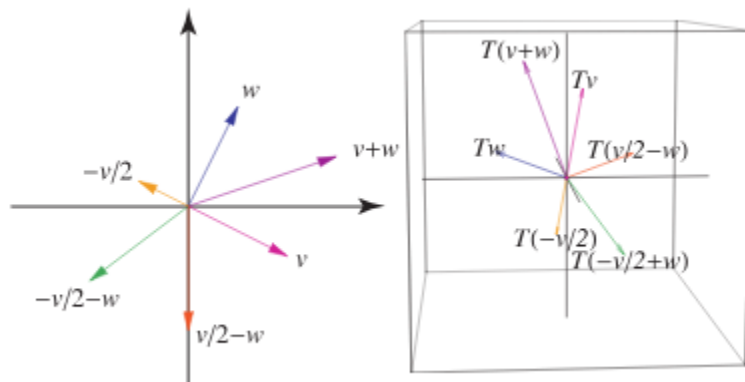
More precisely, a function $T: V \rightarrow W$ is considered a linear transformation if it satisfies the following two properties for any vectors u and v in the vector space V and any scalar c :

1. $T(u + v) = T(u) + T(v)$ (Preservation of addition)
2. $T(cu) = cT(u)$ (Preservation of scalar multiplication)

In simpler terms, a linear transformation takes vectors as inputs and produces vectors as outputs, while preserving vector addition and scalar multiplication.

Linear transformations have several important properties and applications. They are used to study the relationship between vector spaces, analyze systems of linear equations, and solve problems in various fields such as physics, engineering, and computer science.

They can be represented by matrices, and their properties can be analyzed through concepts like null spaces, column spaces, rank, and eigenvectors.



Lecture 22

Linearly Independent Sets; Bases

Linearly Independent Sets; Bases

Let V be an arbitrary nonempty set of objects on which two operations are defined, addition and multiplication by scalars. If the following axioms are satisfied by all objects u, v, w in V and all scalars l and m , then we call V a vector space.

Axioms of Vector Space

For any set of vectors u, v, w in V and scalars l, m, n :

1. $u + v$ is in V
2. $u + v = v + u$
3. $u + (v + w) = (u + v) + w$
4. There exist a zero vector 0 such that $0 + u = u + 0 = u$
5. There exist a vector $-u$ in V such that $-u + u = 0 = u + (-u)$
6. (lu) is in V
7. $l(u + v) = lu + lv$
8. $m(nu) = (mn)u = n(mu)$
9. $(l+m)u = lu + mu$
10. $1u = u$ where 1 is the multiplicative identity

Subspace

A subset W of a vector space V is called a subspace of V if W itself is a vector space under the addition and scalar multiplication defined on V .

Linearly independent, linearly independent set, linearly dependent, linearly dependent set

If the trivial solution is the only solution to this equation then the vectors in the set are called linearly independent and the set is called a linearly independent set. If there is another solution then the vectors in the set are called linearly dependent and the set is called a linearly dependent set.

Useful results

- A set containing the zero vector is linearly dependent.
- A set of two vectors is linearly dependent if and only if one is a multiple of the other.
- A set containing one nonzero vector is linearly independent. i.e. consider the set containing one nonzero vector $\{v_1\}$ so $\{v_1\}$ is linearly independent when $1 \cdot v \neq 0$.
- A set of two vectors is linearly independent if and only if neither of the vectors is a multiple of the other.

The Spanning Set Theorem As we will see, a basis is an “efficient” spanning set that contains no unnecessary vectors. In fact, a basis can be constructed from a spanning set by discarding unneeded vectors.

The Spanning Set Theorem states that for a vector space V , if a set of vectors S can generate or "span" the entire vector space V , then any vector in V can be expressed as a linear combination of the vectors in S .

In other words, if $S = \{v_1, v_2, v_3, \dots, v_n\}$ is a set of vectors in V , and every vector in V can be written as a linear combination of $v_1, v_2, v_3, \dots, v_n$, then S is said to span V .

The Spanning Set Theorem is a fundamental concept in linear algebra as it helps us understand the relationship between vector spaces and their spanning sets. It allows us to determine whether a set of vectors can generate the entire vector space or if it is insufficient.

Additionally, the Spanning Set Theorem is closely related to the concept of linear independence. If a set of vectors spans a vector space and is also linearly independent, it forms a basis for that vector space. A basis is a set of vectors that can generate the entire vector space and has the property that no vector can be expressed as a linear combination of the other vectors in the set. Two Views of a Basis When the Spanning Set Theorem is used, the deletion of vectors from a spanning set must stop when the set becomes linearly independent.